

Table 24: ASV and MR of Combined Attack for different target and injected tasks when random sequences as delimiters is used. The LLM is GPT-4.

Target Task	Injected Task													
	Dup. sentence detection		Grammar correction		Hate detection		Nat. lang. inference		Sentiment analysis		Spam detection		Summarization	
	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR
Dup. sentence detection	0.56	0.65	0.52	0.90	0.29	0.35	0.89	0.89	0.85	0.89	0.08	0.07	0.22	0.37
Grammar correction	0.66	0.75	0.42	0.74	0.54	0.58	0.90	0.93	0.71	0.72	0.54	0.51	0.39	0.65
Hate detection	0.54	0.61	0.46	0.78	0.67	0.81	0.85	0.81	0.94	0.97	0.80	0.75	0.38	0.64
Nat. lang. inference	0.66	0.76	0.56	0.95	0.33	0.33	0.87	0.88	0.85	0.87	0.57	0.58	0.27	0.45
Sentiment analysis	0.56	0.61	0.52	0.91	0.60	0.73	0.90	0.93	0.93	0.94	0.96	0.96	0.45	0.74
Spam detection	0.58	0.63	0.55	0.95	0.71	0.82	0.87	0.92	0.93	0.95	0.83	0.80	0.43	0.71
Summarization	0.71	0.74	0.56	0.93	0.75	0.85	0.88	0.93	0.96	0.98	0.95	0.95	0.42	0.70

Table 25: ASV and MR of Combined Attack for different target and injected tasks when XML tag as delimiters is used. The LLM is GPT-4.

Target Task	Injected Task													
	Dup. sentence detection		Grammar correction		Hate detection		Nat. lang. inference		Sentiment analysis		Spam detection		Summarization	
	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR
Dup. sentence detection	0.62	0.73	0.55	0.93	0.54	0.61	0.96	0.91	0.90	0.94	0.61	0.58	0.43	0.75
Grammar correction	0.77	0.88	0.54	0.92	0.34	0.39	0.91	0.94	0.65	0.69	0.60	0.56	0.41	0.68
Hate detection	0.59	0.67	0.32	0.60	0.63	0.73	0.79	0.79	0.79	0.84	0.55	0.48	0.30	0.52
Nat. lang. inference	0.66	0.75	0.56	0.96	0.38	0.41	0.85	0.86	0.87	0.91	0.79	0.78	0.39	0.70
Sentiment analysis	0.67	0.70	0.55	0.94	0.54	0.72	0.92	0.95	0.96	0.96	0.81	0.79	0.43	0.71
Spam detection	0.58	0.59	0.53	0.93	0.69	0.78	0.89	0.92	0.90	0.92	0.86	0.83	0.43	0.73
Summarization	0.71	0.76	0.56	0.93	0.75	0.89	0.92	0.95	0.94	0.98	0.90	0.87	0.42	0.73

Table 26: ASV and MR of Combined Attack for different target and injected tasks when instructional prevention is used. The LLM is GPT-4.

Target Task	Injected Task													
	Dup. sentence detection		Grammar correction		Hate detection		Nat. lang. inference		Sentiment analysis		Spam detection		Summarization	
	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR
Dup. sentence detection	0.53	0.64	0.32	0.55	0.03	0.05	0.14	0.13	0.12	0.13	0.02	0.02	0.01	0.02
Grammar correction	0.71	0.86	0.57	0.96	0.42	0.43	0.55	0.57	0.45	0.48	0.18	0.16	0.28	0.40
Hate detection	0.01	0.01	0.15	0.32	0.60	0.71	0.02	0.03	0.10	0.11	0.02	0.02	0.01	0.01
Nat. lang. inference	0.44	0.53	0.51	0.93	0.28	0.32	0.74	0.77	0.68	0.72	0.38	0.37	0.14	0.25
Sentiment analysis	0.46	0.46	0.42	0.79	0.49	0.58	0.59	0.61	0.89	0.93	0.32	0.34	0.18	0.30
Spam detection	0.14	0.14	0.46	0.86	0.20	0.22	0.34	0.35	0.21	0.22	0.60	0.55	0.03	0.05
Summarization	0.71	0.78	0.54	0.92	0.73	0.80	0.89	0.94	0.95	0.97	0.91	0.88	0.38	0.65

Table 27: ASV and MR of Combined Attack for different target and injected tasks when sandwich prevention is used. The LLM is GPT-4.

Target Task	Injected Task													
	Dup. sentence detection		Grammar correction		Hate detection		Nat. lang. inference		Sentiment analysis		Spam detection		Summarization	
	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR
Dup. sentence detection	0.61	0.72	0.04	0.07	0.06	0.07	0.77	0.74	0.37	0.38	0.74	0.74	0.12	0.21
Grammar correction	0.35	0.39	0.49	0.85	0.21	0.20	0.55	0.57	0.13	0.14	0.05	0.04	0.04	0.04
Hate detection	0.27	0.29	0.03	0.11	0.69	0.80	0.41	0.41	0.89	0.92	0.21	0.21	0.00	0.00
Nat. lang. inference	0.70	0.79	0.51	0.88	0.49	0.58	0.86	0.87	0.89	0.91	0.76	0.77	0.31	0.54
Sentiment analysis	0.24	0.29	0.04	0.29	0.35	0.41	0.14	0.13	0.97	0.97	0.08	0.08	0.01	0.00
Spam detection	0.53	0.52	0.13	0.46	0.54	0.58	0.91	0.90	0.93	0.95	0.91	0.88	0.05	0.08
Summarization	0.66	0.83	0.50	0.88	0.65	0.65	0.86	0.91	0.94	0.98	0.89	0.86	0.43	0.73

Table 28: FNR of PPL detection at detecting Combined Attack for different target and injected tasks. The LLM is Llama-2-13b-chat.

Target Task	Injected Task						
	Dup. sentence detection	Grammar correction	Hate detection	Nat. lang. inference	Sentiment analysis	Spam detection	Summarization
Dup. sentence detection	1.00	0.74	0.61	1.00	0.59	0.54	0.92
Grammar correction	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Hate detection	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Nat. lang. inference	1.00	0.86	0.65	1.00	0.71	0.66	0.96
Sentiment analysis	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Spam detection	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Summarization	1.00	0.99	0.94	1.00	0.92	0.91	1.00

Table 29: FNR of windowed PPL detection at detecting Combined Attack for different target and injected tasks. The LLM is Llama-2-13b-chat.

Target Task	Injected Task						
	Dup. sentence detection	Grammar correction	Hate detection	Nat. lang. inference	Sentiment analysis	Spam detection	Summarization
Dup. sentence detection	0.85	0.26	0.09	0.79	0.09	0.10	0.61
Grammar correction	1.00	0.99	0.98	1.00	1.00	0.98	1.00
Hate detection	1.00	0.99	0.99	1.00	1.00	0.98	1.00
Nat. lang. inference	0.95	0.31	0.31	0.98	0.34	0.25	0.87
Sentiment analysis	0.97	0.93	0.90	0.95	0.99	0.88	0.99
Spam detection	1.00	0.98	0.97	1.00	1.00	0.96	0.99
Summarization	0.93	0.78	0.57	0.92	0.66	0.45	0.97

Table 30: FNR of naive LLM-based detection at detecting Combined Attack for different target and injected tasks. The LLM is GPT-4.

Target Task	Injected Task						
	Dup. sentence detection	Grammar correction	Hate detection	Nat. lang. inference	Sentiment analysis	Spam detection	Summarization
Dup. sentence detection	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Grammar correction	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Hate detection	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Nat. lang. inference	0.00	0.00	0.01	0.00	0.00	0.00	0.00
Sentiment analysis	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Spam detection	0.00	0.00	0.00	0.00	0.01	0.00	0.00
Summarization	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 31: FNR of response-based detection at detecting Combined Attack for different target and injected tasks. The LLM is GPT-4.

Target Task	Injected Task						
	Dup. sentence detection	Grammar correction	Hate detection	Nat. lang. inference	Sentiment analysis	Spam detection	Summarization
Dup. sentence detection	1.00	0.06	0.02	0.00	0.00	0.02	0.00
Grammar correction	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Hate detection	0.05	0.02	0.87	0.03	0.00	0.05	0.00
Nat. lang. inference	0.00	0.06	0.03	1.00	0.00	0.01	0.01
Sentiment analysis	0.00	0.02	0.01	0.00	1.00	0.07	0.00
Spam detection	0.00	0.22	0.01	0.00	0.00	0.95	0.02
Summarization	1.00	1.00	1.00	1.00	1.00	1.00	1.00